

Insights

Milestone 1:- The dataset contains 46,185 disaster records and covers 59 unique states/regions, showing wide geographic coverage. After cleaning, key columns like state and incident type have no missing values, ensuring reliable analysis. Some fields like county and end date have minimal missing data, while close date has a significant number of missing values, indicating many cases are still open or not updated. Time-based features such as year and month were extracted, enabling trend and seasonal analysis. Overall, the dataset is well-structured and ready for further analysis and visualization.

Milestone 2:- The analysis shows a clear **increasing trend in disasters over the years**, indicating that natural disasters are becoming more frequent. Among different types, **floods and storms occur most frequently**, making them the dominant disaster categories. The monthly analysis reveals strong **seasonality**, with higher occurrences during specific months due to events like hurricanes, monsoons, and wildfires. Peak years such as **2005, 2011, and 2008** experienced exceptionally high disaster counts, likely due to multiple major events. Overall, the rise in disasters can be linked to **climate change, extreme weather conditions, and improved reporting systems**.

Milestone 3:- The analysis clearly shows that **storms, floods, and hurricanes are the most frequent disasters across the dataset**. Among them, **storms have the highest occurrence**, contributing the largest share in most states (for example, Alabama has around **529 storm events**, Arkansas about **603 storms**, and Alaska around **64 storms**). **Floods are the second most common disaster**, with states like Arkansas (**235 floods**), Alabama (**104 floods**), and Alaska (**46 floods**) showing high numbers. **Hurricanes also contribute significantly**, especially in southern states like Alabama (**368 hurricanes**) and Arkansas (**93 hurricanes**). In contrast, disasters like earthquakes, tsunamis, chemical incidents, and terrorism occur very rarely. Overall, the data indicates that **weather-related disasters (storms, floods, hurricanes)** dominate, making them the most impactful and frequent disaster types in the United States.

Milestone 4:- The analysis shows that **storms are the most frequent disaster**, with a total of **16,250 occurrences**, followed by **floods (9,317)** and **hurricanes (8,764)**, indicating that weather-related disasters dominate the dataset. These disasters mainly occur in **southern and coastal states** such as Texas, Florida, and Louisiana, where climatic conditions favor hurricanes, heavy rainfall, and storms. Inland states tend to experience more **floods and tornadoes**, while colder regions show higher occurrences of **snow and ice-related disasters**. Additionally, these high-frequency disasters (storms, floods, hurricanes) also trigger the maximum number of **assistance programs**, proving that they have the greatest impact on both people and infrastructure. Overall, the insights highlight that **geography and climate play a major role in determining the type and frequency of disasters**.